MEDSURE : HACKATHON LIVE DEMO SCRIPT

Tagline: Know the Fair Price. Verify Every Bill. | ■■■ ■■■ ■■■■■ ■■ ■■■ ■■■■■■

CORE ARCHITECTURAL THESIS: *'AI understands language and vision; deterministic rules decide the finances.'* Unlike ungrounded LLM wrappers that hallucinate prices or medical advice, MEDSURE separates linguistic comprehension and OCR parsing from mathematical audit calculations. All financials are computed with Python Decimal precision.

1. DEMONSTRATION RUNWAY (3-MINUTE SCHEDULE)

Time	Act & Target	On-Screen Action / Route	Core Takeaway
0:00 - 0:30	The Hook & Vision	Homepage (/) & Dashboard (/dashboard)	60%+ out-of-pocket Indian healthcare burden; LLMs fail at math.
0:30 - 1:15	Act 1: Before Admission	Triage (/triage) with Hindi preset	Safety triage passed; Knee TKR bounds: Min 1.25L, Median 1.45L, Cap 1.85L.
1:15 - 2:10	Act 2: Medical Bill OCR & Audit	Synthetic Bill -> Review (/bills/[id]/review)	Deterministic Rules A-E fire live: catches **Rs. 92,500** in overcharges!
2:10 - 2:40	Resolution & Patient Power	PDF Report & Dispute Letter modal	Official ReportLab PDF report & 1-click polite hospital dispute letter.
2:40 - 3:00	Tech Stack & Wrap-up	Dashboard (/dashboard) & Final pitch	FastAPI, PyMuPDF, ReportLab, Next.js, Decimal math, 100% test pass rate.

2. VERBATIM STAGE SCRIPT & ACTION CUES

ACT 0: THE HOOK (0:00 - 0:30)

ACTION CUE: Presenter 1 on Homepage (http://localhost:3000), navigating smoothly to /dashboard.

"Good morning judges. In India, over 60% of all healthcare expenses come directly out of a patient's personal life savings. When a family member is hospitalized, patients are vulnerable, grieving, and exhausted. They are handed an 8-page invoice packed with cryptic abbreviations and a 4-lakh total—and they pay it because they have zero leverage.

If you dump that bill into ChatGPT, it hallucinates rates and invents medical facts. That is unacceptable in healthcare.

Meet MEDSURE. Our core principle is: AI understands language and vision, but deterministic code does the math. Every single calculation is verified with zero LLM rate hallucination."

ACT 1: BEFORE ADMISSION — FAIR PRICE ESTIMATOR (0:30 - 1:15)

ACTION CUE: Click 'Fair Price Estimator' (/triage). Click preset 'Orthopedic Knee Pain (Hindi)'. Click 'Calculate Fair Price Range'.

"MEDSURE protects patients across two critical moments: Before Admission and After Discharge.

Let's look at Before Admission. Most Indian families search for care in their mother tongue. We enter symptoms in colloquial Hindi: '■■■■ ■■■■■ ■■■■ ■■■■ ■■■ ■■■■■ ■■...'

Notice what MEDSURE does first: it runs a Clinical Safety Screen. If the patient entered acute chest pain or sudden paralysis, pricing is instantly suppressed to route them to emergency 112/108 services.

Because this query is non-emergency, it maps to Orthopedics — Total Knee Replacement, and extracts official price caps from CGHS and NPPA benchmarks. Patients see transparent bounds on an interactive chart: Minimum Rs. 1,25,000, Median Rs. 1,45,000, and a Ceiling of Rs. 1,85,000, with clear inclusions and exclusions. Now the patient enters the hospital informed, not intimidated."

ACT 2: AFTER DISCHARGE — OCR & DETERMINISTIC AUDIT (1:15 - 2:10)

ACTION CUE: Navigate to /bills/upload. Click 'Load Synthetic Bill Demo' (Apex Care Hospital Knee Surgery bill). Show /bills/[id]/review. Click 'Run Deterministic Audit'.

"Now comes Act 2: Discharge day. The patient receives an itemized invoice. We load it into MEDSURE.

Using PyMuPDF and OCR vision, it parses every single line item into structured data with confidence scores. Our Human-in-the-Loop review screen lets patients correct any misread character before financial audit.

Now watch: we click 'Run Deterministic Audit'. In less than 300 milliseconds, our Python engine evaluates Rules A through E with exact Decimal precision. Look at what it caught:
• Rule A (Arithmetic): Surgeon Operating Fee—1 × Rs. 70,000 billed as Rs. 85,000. That's a direct Rs. 15,000 math discrepancy!
• Rule B (Duplicate Charge): Operation Theatre Fee charged twice in a single surgical session. Discrepancy: Rs. 35,000.
• Rule C (Price Cap Breach): Knee Implant Prosthesis billed at Rs. 1,15,000 against the NPPA statutory ceiling of Rs. 85,000. Discrepancy: Rs. 30,000.
• Rule D (Unbundling): Routine surgical drapes billed on top of an all-inclusive surgery package.

Total potential unjustified overcharge identified: Rs. 92,500.00! Every finding provides exact mathematical proof and regulatory evidence."

ACT 3: THE RESOLUTION — PDF REPORT & DISPUTE LETTER (2:10 - 2:40)

ACTION CUE: Click 'Download Official PDF Report' -> Opens ReportLab PDF. Then click 'Generate Hospital Dispute Letter' -> Modal opens.

"Catching overcharges is only half the victory. Arming the patient to resolve them is the cure.

First, MEDSURE generates an official signed PDF Audit Report using ReportLab. It contains all line-item breakdowns, audit findings, and statutory references formatted for the hospital billing desk or insurance TPA.

Second, click 'Generate Hospital Dispute Letter'. MEDSURE crafts an editable, respectful, legally compliant dispute letter addressed directly to the Billing Desk and Chief Medical Superintendent, citing exact line items and requesting an itemized reconciliation. One click copies it to clipboard or emails it."

CONCLUSION & WRAP-UP (2:40 - 3:00)

ACTION CUE: Switch back to Dashboard (/dashboard) showing updated system metrics. Make strong eye contact with judges.

"Under the hood: Next.js 14 App Router, FastAPI backend, pure Python Decimal arithmetic, PyMuPDF OCR processing, and 100% automated test coverage.

Healthcare billing should never be a black box that exploits families at their lowest moments. With MEDSURE, every patient can Know the Fair Price, and Verify Every Bill. Thank you!"

3. JUDGES Q&A; CHEAT SHEET (INSTANT WINNING RESPONSES)

Judge Question	Winning Answer (What to say)
'Why not just upload the bill to ChatGPT or Claude?'	LLMs are probabilistic token predictors, not arithmetic engines. They frequently miscalculate sums, hallucinate non-existent price caps, and invent medical rules. In healthcare finance, an ungrounded LLM creates catastrophic liability. MedSure uses NLP strictly for language parsing; all audit math is executed deterministically with Decimal precision.
'Where do your benchmark rates come from?'	In our production roadmap, MedSure aggregates verified public sources: Central Government Health Scheme (CGHS) scheduled rates, National Pharmaceutical Pricing Authority (NPPA) ceiling caps on cardiac stents and orthopedic implants, and ROHINI hospital registry data. Unlisted procedures use transparent empirical median ranges.
'What if OCR misreads a crumpled or handwritten invoice?'	We built a Human-in-the-Loop review screen (/bills/[id]/review). Extracted line items display confidence scores. Patients can verify and edit quantities or amounts before the audit engine runs. We never execute automated financial verdicts on raw unverified OCR output.
'Is this legal advice or hospital confrontation?'	No. MedSure is a transparency tool. Every screen carries clinical and legal disclaimers. Our generated dispute letter is polite, evidence-backed, and framed as an itemized reconciliation request, which billing desks appreciate because it saves them hours of manual recalculation.
'What is the monetization / business model?'	1. B2C: Free basic scan with premium deep dispute support. 2. B2B Insurance & TPAs: Pre-adjudication engine to detect fraudulent and inflated hospital claims before disbursal, saving health insurers billions. 3. Corporate wellness benefit integration.